

of manufacturing makes wastage near negligible. In industries where metal is used, it is found that traditional methods leave nearly 90 percent of the source apart from the final product due to finishing and polishing steps, which creates more waste per product. With modern metal based 3D printers, this wastage is significantly minimised making 3D printing a far more eco friendly alternative.

Industrial use, prototyping and efficiency

All great innovations require large scale investments to find footing as they evolve. In this respect, 3D printing is already seeing significant interest from corporations and investors. The large scale usage of 3D printing not only makes it a viable alternative to traditional manufacturing methods and an alluring investment opportunity for innovators but also allows it to be integrated into the mainstream consumer usage. The industry in this respect is critical towards making 3D printing a norm in society.

One of the fundamental methods of industrial usage for 3D printing lies in its use for rapid prototyping. The range of techniques that encompass rapid prototyping allow large scale manufacturers to very quickly create a simulated fabrication of any physical object in a three dimensional computer aided program. As industries use 3D printing to produce and test their products it brings the cost of creating mass producible items down and makes it cheaper for the end-user.

Rapid prototyping through 3D printing also enables industries to create the exact number of required parts without relying on the economies of scale, which would otherwise force them to expend resources on larger manufacturing orders. The print to use principle of 3D printing becomes an integral component of product design and costing.

Currently the use of 3D printing in rapid prototyping is in its infancy but has been described as the “next level” of manufacturing technology since 2009 by many experts. The key obstacle in the use of 3D printers within industry lies in its relatively slow speed of printing. This barrier is being



A car engine prototype can be printed in hours and ready for testing. Faster than ever before and cheaper too.